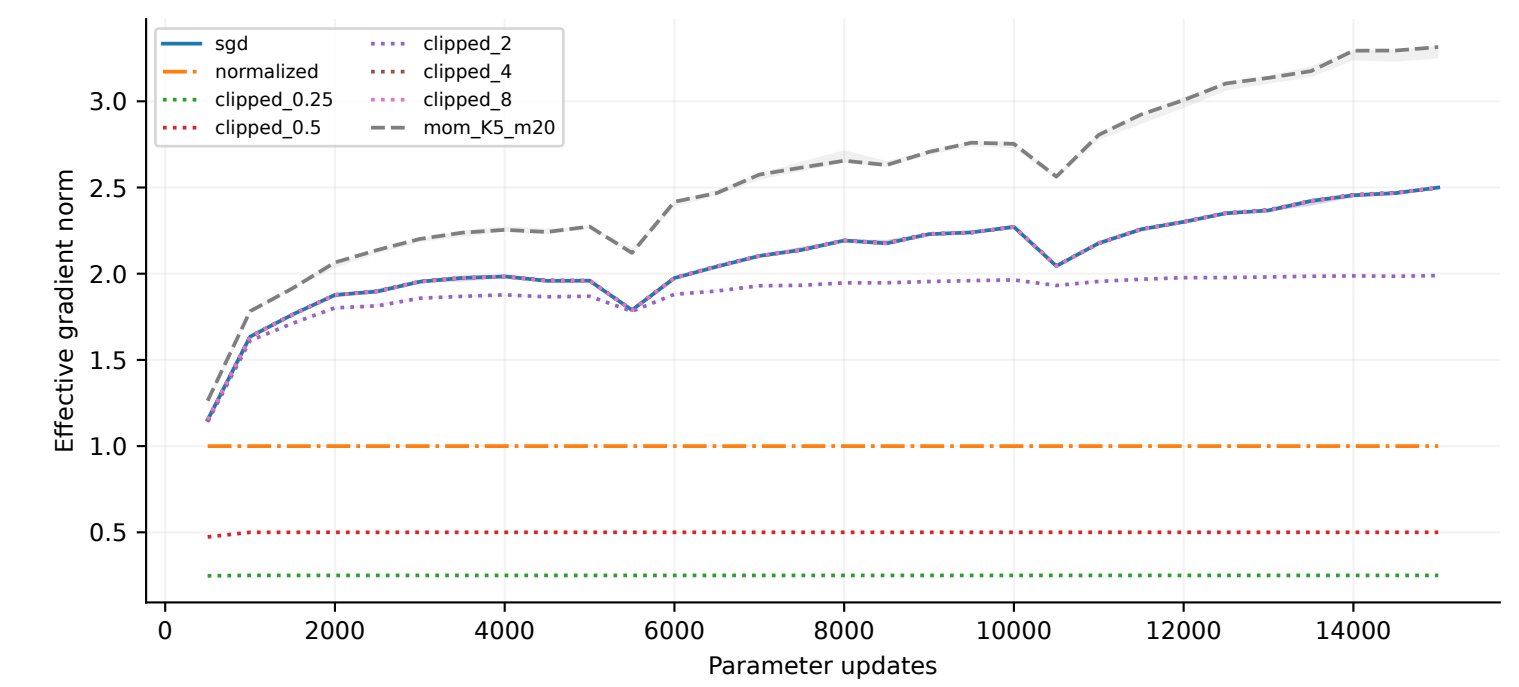
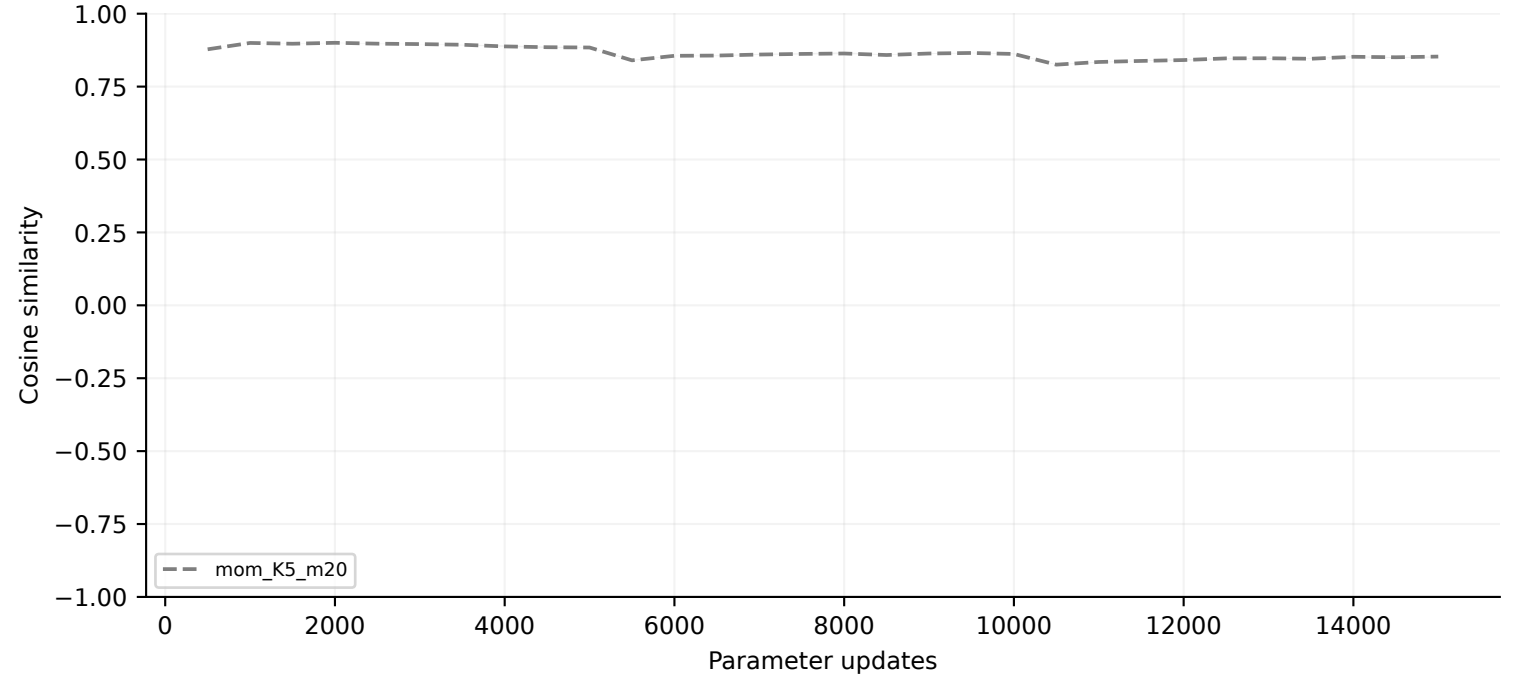
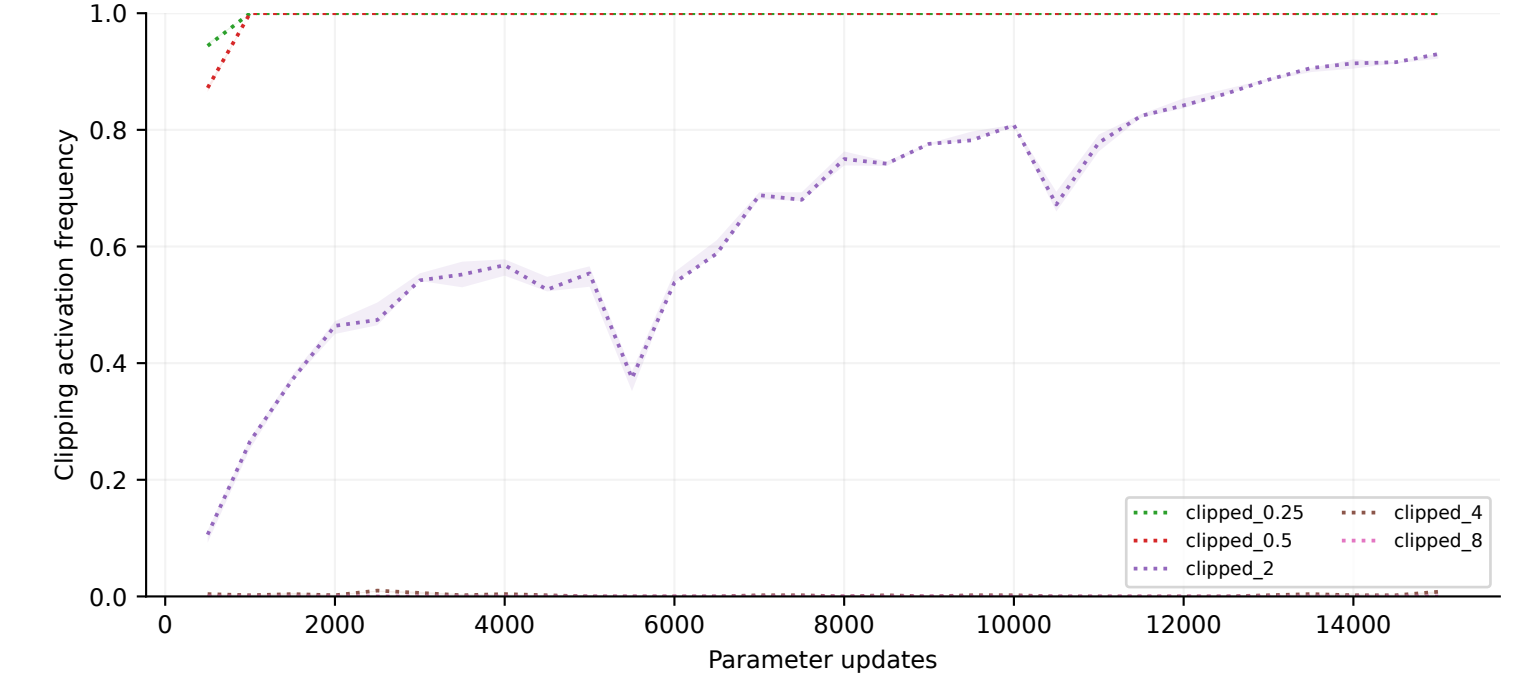

sgd
normalized
clipped_0.25
clipped_0.5
clipped_2
clipped_4
clipped_8
mom_K5_m20
Effective gradient norm
3.0
2.5
2.0
1.5
1.0
0.5
0
2000
4000
6000
8000
10000
12000
14000
Parameter updates


Cosine similarity
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
mom_K5_m20
0
2000
4000
6000
8000
10000
12000
14000
Parameter updates


Clipping activation frequency
1.0
0.8
0.6
0.4
0.2
0.0
clipped_0.25
clipped_0.5
clipped_2
clipped_4
clipped_8
0
2000
4000
6000
8000
10000
12000
14000
Parameter updates